Notes on Normal forms of cocycles and regular neighborhoods

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Some long-term characteristics of a dynamical system, including Lyapunov exponents, Oseledets splitting, LP regularity, etc, can be studied as the action of $\mathrm{d}f$ on TM. We can even study a more general formulation on any vector bundle, the cocycle. The results below belong to ergodic theory, where we use measure theory to study dynamical systems, so throughout, we work on measure spaces $(X, \mathcal{B}, \mu)$, and $f : X \to X$ is an invertible and measurable map.

In the first section, we introduce our objects, maps, and equivalences. In Section 2, we aim to explain a normal form for cocycles. The significance of this is that certain dynamical information is nicely preserved along the entire trajectory. And in section 3, we will apply this to $\mathrm{d}f$ to get an Anosov neighborhood around any LP regular point.

1 General Setups

Definition 1.1 (Cocycle)**.** A **cocycle** $\mathcal{A}$ is a matrix-valued function generated by a sequence of matrices

$$\mathcal{A}(m, l) = A_{m+l-1} \cdots A_m$$

that satisfies the cocycle condition:

$$\mathcal{A}(m, l + k) = \mathcal{A}(m + k, l)\mathcal{A}(m, k).$$

This naturally generalizes to **linear multiplicative cocycle**, which encodes the information of f.

Definition 1.2 (Linear multiplicative cocycle)**.** Given $f : X \to X$ a measurable transformation, with a function $\mathcal{A} : X \times \mathbb{Z} \to \mathrm{GL}_n(\mathbb{R})$ such that

(1) For each x, $\mathcal{A}(x, 0) = \mathrm{Id}$, and $\mathcal{A}(x, m + k) = \mathcal{A}(f^k(x), m)\mathcal{A}(x, k)$. Clearly, $A(x) = \mathcal{A}(x, 1)$ **generates** the cocycle.

(2) For all $m \in \mathbb{Z}$, $x \mapsto \mathcal{A}(x, m)$ is measurable.

$\mathcal{A}$ induces a natural **linear extension** on the trivial bundle $F : X \times \mathbb{R}^n \to X \times \mathbb{R}^n$, given by $(x, v) \mapsto (f(x), A(x)v)$.

Consider a diffeomorphism of a compact smooth Riemannian n-manifold M, then given a trajectory $\{f^m(x)\}_{m\in\mathbb{Z}}$, we can identify each $T_{f^m(x)}M$ with $\mathbb{R}^n$ and thus obtain a sequence of matrices $A_m(x) \coloneqq \mathrm{d}f_{f^m(x)}$. This is referred to as the **individual derivative cocycle** associated with x.

Remark: These are *not* representations of $\langle f\rangle \leq \mathrm{Aut}(X)$. However, if $f(x) = x$, there is something to say (not here).

Linear extensions are bundle maps between measurable vector bundles. Let $\pi : E \to X$, E is called a **measurable vector bundle** over X if there is a countable collection of measurable subsets $X_i \subset X$ which cover X, and measurable maps $\pi_i : X_i \times \mathbb{R}^n \to X$. Measure theoretically, this is general for any compact metric space. Indeed, we have:

Proposition 1.1. Let (X, μ) be a compact metric space. If $\pi : E \to X$ is a measurable vector bundle over field k, then there is a full measure subset $Y \subset X$ such that $\pi^{-1}(Y) \cong Y \times k^n$.

This actually works for a σ-compact metric space. Anyway, we may assume the bundle is trivial mod 0.
We proceed with an equivalence relation on cocycles. Two cocycles $\mathcal{A}$ and $\mathcal{B}$ are said to be **tempered equivalent** on $Y \subset X$, where Y is an f-invariant, nonempty, measurable, and satisfies

$$A(x) = L(f(x))^{-1}B(x)L(x),$$

where $L : X \to \mathrm{GL}(n, \mathbb{R})$ is **tempered** on Y, i.e. for every $x \in Y$, we have

$$\lim_{m\to\pm\infty} \frac{1}{m} \log \|L^{\pm 1}(f^m(x))\| = 0. \tag{1}$$

Theorem 1 (Multiplicative Ergodic Theorem)**.** Assume that the generator A of the cocycle $\mathcal{A}$ satisfies the **integrability condition**, i.e. $\log^+ \|A^{\pm}\| \in L^1(X, \nu)$, then there are ν-a.e. LP-regular points for $\mathcal{A}$.

The proof is in the textbook and is too long to put here. Nevertheless, we note that in the proof we reduce the general case to upper triangular cocycles, which questions whether we can somehow reduce the general cases to a simpler case via some "conjugation". It turns out that effort had been made in this direction.

We introduce the normal form of cocycles.

2 Normal forms of cocycles

The goal of this section is to describe a normal form over $f : X \times \mathbb{R}^n \to X \times \mathbb{R}^n$ associated with its Lyapunov exponents. We want a normal form that should be reduced to the Jordan normal form when the generator is a constant matrix. A cocycle satisfying the integrability condition can be reduced to a constant cocycle up to an arbitrarily small error. We are essentially using the tools of analysis, which add extra rigidity, to study an algebraic structure.

Proposition/Definition 2.1. For every $\varepsilon > 0$, each LP-regular point $x \in X$, each $i = 1, \dots, s(x)$ in the Oseledets decomposition, the following is an inner product *at* x, and is called **strong Lyapunov inner product**,

$$\langle u, v\rangle'_x = \sum_{i=1}^{s(x)} \sum_{m\in\mathbb{Z}} \langle \mathcal{A}(x,m)u_i, \mathcal{A}(x,m)v_i\rangle e^{-2\chi_i(x)m-2\varepsilon|m|}.$$

The inner product is not necessarily continuous on X. The inner product is defined by a limit of measurable functions, hence measurable. Also note that the weight $e^{-2\chi_i(x)m-2\varepsilon|m|}$ ensures the convergence for any $\varepsilon > 0$.

Definition 2.1 (Lyapunov change of coordinates)**.** A coordinate change $C_\varepsilon : X \to \mathrm{GL}(n, \mathbb{R})$ is called a **Lyapunov change of coordinates** if for each LP-regular point $x \in X$, it is an isometry: $\langle u, v\rangle = \langle C_\varepsilon(x)u, C_\varepsilon(x)v\rangle'_x$. Any such C_ε sends the orthogonal decomposition $\bigoplus_{i=1}^{s(x)} \mathbb{R}^{k_i(x)}$ to $\bigoplus_{i=1}^{s(x)} E_i(x)$, the Oseledets decomposition, where $k_i(x) = \dim E_i(x)$.

What is explicitly given below states that we can conjugate $A(x)$ to a block form while preserving dynamical properties.

Theorem 2 (Oseledets-Pesin Reduction Theorem)**.** Let $f : X \to X$ be an invertible measure-preserving transformation of the Lebesgue space (X, ν), and let $\mathcal{A}$ be a cocycle over f. Given $\varepsilon > 0$, if x is an LP-regular point for $\mathcal{A}$, then

(i) The generator $A_\varepsilon(x) = C_\varepsilon(f(x))^{-1}A(x)C_\varepsilon(x)$ for the cocycle $\mathcal{A}_\varepsilon(x, m)$ has the block form

$$A_\varepsilon(x) = \begin{pmatrix} A_\varepsilon^1(x) & & \\ & \ddots & \\ & & A_\varepsilon^{s(x)}(x) \end{pmatrix}$$

where each block is of size $k_i(x) \times k_i(x)$.

(ii) It holds that

$$e^{\chi_i(x)-\varepsilon} \le \|A_\varepsilon^i(x)^{-1}\|^{-1} \le \|A_\varepsilon^i(x)\| \le e^{\chi_i(x)+\varepsilon}.$$

(iii) If the generator A of the cocycle $\mathcal{A}$ satisfies the integrability condition, then for ν-almost every $x \in X$, the map C_ε is tempered and the Lyapunov spectra of $\mathcal{A}$ and $\mathcal{A}_\varepsilon$ coincide.

Proof. (i). It suffices to verify the invariance of $\mathbb{R}^{k_i(x)}$:

$$A_\varepsilon(x)\mathbb{R}^{k_i(x)} = C_\varepsilon(f(x))^{-1}A(x)C_\varepsilon(x)\mathbb{R}^{k_i(x)} = C_\varepsilon(f(x))^{-1}A(x)E_i(x) = C_\varepsilon(f(x))^{-1}E_i(f(x)) = \mathbb{R}^{k_i(f(x))} = \mathbb{R}^{k_i(x)}.$$

(ii). If $v = C_\varepsilon(x)w$, then $\|w\| = \|v\|'$, and $\|A_\varepsilon^i(x)w\| = \|A^i(x)v\|'_{f(x)}$. It suffices to estimate for $\|\cdot\|'$. We have the identity:

$$(\|A(x)v\|'_{f(x)})^2 = \sum_{m\in\mathbb{Z}} \|\mathcal{A}(x, m+1)v\|^2 e^{-2m\chi_i(x)-2\varepsilon|m|} = (\|v\|'_x)^2 e^{2\chi_i(x)+2\varepsilon(|m|-|m-1|)}.$$

Since $||m| - |m-1|| = 1$, we have, for every $v \in E_i(x)$, $\|v\|'_x e^{\chi_i(x)-\varepsilon} \le \|A(x)v\|'_{f(x)} \le \|v\|'_x e^{\chi_i(x)+\varepsilon}$.

(iii). We estimate the operator norms. For every $w \in \mathbb{R}^n$, LP-regular point x, $\|w\| = \|C_\varepsilon(x)w\|'_x \ge \|C_\varepsilon(x)w\|$, hence

$$\|C_\varepsilon(x)\| \le 1, \text{ and } \|C_\varepsilon^{-1}(x)\|^{-1} \le 1. \tag{2}$$

Define for $N \in \mathbb{N}$ the set

$$X_N = \{x \in X \mid x \text{ is LP-regular}, \|C_\varepsilon(x)\| > N^{-1}, \|C_\varepsilon^{-1}(x)\|^{-1} > N^{-1}\} \subset X_{N+1}.$$

By Theorem 1, we have $\nu(\bigcup_{N\ge1} X_N) = 1$, implying that there is an N so that $\nu(X_N) > 0$. Choose the full measure Poincaré recurrent subset $Y_N \subset X_N$ in which all points return infinitely many times. For each $x \in Y_N$, denote the return time by $m_k = m(k, x)$, which is the sequence of positive (resp. negative) integers so that $f^{m_k}(x) \in Y_N$.

For each $i = 1, \dots, s(x)$, since $\mathcal{A}_\varepsilon(x, m) = C_\varepsilon^{-1}(f^m(x))\mathcal{A}(x, m)C_\varepsilon(x)$, we have the following two estimates:

$$\|\mathcal{A}_\varepsilon^i(x, m_k)\| = \|C_\varepsilon^{-1}(f^{m_k}(x))\|\|\mathcal{A}(x, m_k)|E_i(x)\|\|C_\varepsilon(x)\| \le N\|\mathcal{A}(x, m_k)|E_i(x)\|; \tag{3}$$

$$\|\mathcal{A}(x, m_k)|E_i(x)\| = \|C_\varepsilon(f^{m_k}(x))\|\|\mathcal{A}_\varepsilon^i(x, m_k)\|\|C_\varepsilon^{-1}(x)\| \le N\|\mathcal{A}_\varepsilon^i(x, m_k)\|. \tag{4}$$

Taking lim sup to both sides of (3), and lim inf to (4) yields (resp. $m \to -\infty$)

$$\limsup_{m\to+\infty} \frac{1}{m} \log \|A_\varepsilon^i(x, m)\| \le \chi_i(x) \le \liminf_{m\to+\infty} \frac{1}{m} \log \|A_\varepsilon^i(x, m)\| = \lim_{m\to+\infty} \frac{1}{m} \log \|A_\varepsilon^i(x, m)\|. \tag{5}$$

This shows that the Lyapunov spectra of $\mathcal{A}$ and $\mathcal{A}_\varepsilon$ coincide for all N, hence ν-almost everywhere.

We now show that $C_\varepsilon(f^m(x))$ is tempered. Observe that

$$1 \le \|C_\varepsilon^{-1}(f^m(x))\| = \|\mathcal{A}_\varepsilon(x,m)C_\varepsilon^{-1}(x)\mathcal{A}(x,m)^{-1}\| = \max_i \sup_{w\in E_i(x)} \frac{\|\mathcal{A}_\varepsilon(x,m)C_\varepsilon^{-1}(x)w\|}{\|\mathcal{A}(x,m)w\|},$$

and that

$$\max_i \sup_{w\in\mathbb{R}^{k_i(f^m(x))}} \frac{\|\mathcal{A}(x,m)C_\varepsilon^{-1}(x)w\|}{\|\mathcal{A}_\varepsilon(x,m)w\|} = \|\mathcal{A}(x,m)C_\varepsilon(x)\mathcal{A}_\varepsilon^{-1}(x,m)\| = \|C_\varepsilon(f^m(x))\| \le 1.$$

Using (2) and (5), we have, for ν-almost every LP-regular $x \in X$,

$$0 \le \liminf_{m\to\pm\infty} \frac{1}{|m|}\log\|C_\varepsilon^{-1}(f^m(x))\| \le \limsup_{m\to\pm\infty} \frac{1}{|m|}\|C_\varepsilon^{-1}(f^m(x))\| = \max_i \sup_{w\in E_i} \lim_{m\to\pm\infty} \frac{1}{|m|}\log\frac{\|\mathcal{A}_\varepsilon(x,m)C_\varepsilon^{-1}(x)w\|}{\|\mathcal{A}(x,m)w\|} = 0,$$

and

$$0 \ge \limsup_{m\to\pm\infty} \frac{1}{|m|}\log\|C_\varepsilon(f^m(x))\| \ge \liminf_{m\to\pm\infty} \frac{1}{|m|}\|C_\varepsilon(f^m(x))\| = \max_i \sup_{w\in\mathbb{R}^{k_i(x)}} \lim_{m\to\pm\infty} \frac{1}{|m|}\log\frac{\|\mathcal{A}(x,m)C_\varepsilon^{-1}(x)w\|}{\|\mathcal{A}_\varepsilon(x,m)w\|} = 0.$$

These imply

$$\lim_{m\to\pm\infty} \frac{1}{|m|}\log\|C_\varepsilon(f^m(x))\| = \lim_{m\to\pm\infty} \frac{1}{|m|}\log\|C_\varepsilon^{-1}(f^m(x))\| = 0,$$

which proves that C_ε is tempered. □

This theorem gives a sufficient condition under which a cocycle can be tempered equivalent to a constant cocycle determined completely by the Oseledets decomposition.

3 Regular Neighborhood

Theorem 2 applies to any cocycle over a measure-preserving invertible map f. In particular, it applies to the derivative cocycle $\mathrm{d}f$ of a smooth measure-preserving diffeomorphism f over a compact Riemannian manifold M. However, the diffeomorphism condition is much stronger than merely measure-preserving and invertible. Let's say f is a diffeomorphism, then $\mathrm{d}f$ is at least continuous. Maybe in this case we can extend the conclusion of Theorem 2 to a chart of an LP-regular point x? The answer is that we need $f \in C^{1+\alpha}$ for the guess to be true.

So our next goal is to construct a neighborhood $N(x) \ni x$ for every LP-regular x such that f acts in $N(x)$ very much like $A_\varepsilon(x)$ in a neighborhood of the origin. This way, f is **uniformly hyperbolic** in $N(x)$ if there's no zero Lyapunov exponent.

To this end, we state a proposition without proving it.

Lemma 3.1 (Tempering Kernel Lemma)**.** If $K : X \to \mathbb{R}$ is a positive measurable function *tempered* on some subset $Z \subset X$, then for any $\varepsilon > 0$, there is a positive measurable function $K_\varepsilon : Z \to \mathbb{R}$ such that $K(x) \le K_\varepsilon(x)$ and for $x \in Z$,

$$K_\varepsilon(x)e^{-\varepsilon} \le K_\varepsilon(f(x)) \le K_\varepsilon(x)e^{\varepsilon}.$$

Such a function K_ε is called a **tempering kernel**.

Theorem 3. Let $\mathcal{R}$ be the set of all LP-regular points of $f \in C^{1+\alpha}$. For every $\varepsilon > 0$, any $x \in \mathcal{R}$, the following hold:

(1) There exists a tempered function $q : \mathcal{R} \to (0,1]$ and a collection of embedding $\Psi_x : B(0,q(x)) \to M$ for each $x \in \mathcal{R}$ such that $\Psi_x(0) = x$ and $e^{-\varepsilon} < \frac{q(f(x))}{q(x)} < e^{\varepsilon}$. These embeddings satisfy $\Psi_x = \exp_x \circ C_\varepsilon(x)$.

(2) If $f_x := \Psi_{f(x)}^{-1} \circ f \circ \Psi(x) : B(0,q(x)) \to \mathbb{R}^p$, then $\mathrm{d}_0 f_x$ has the Lyapunov block form as is in (i) of Theorem 2.

(3) $d_{C^1}(f_x, \mathrm{d}_0 f_x) < \varepsilon$ in $B(0,q(x))$.

(4) There exists a constant $K > 0$ and a measurable function $A : \mathcal{R} \to \mathbb{R}$ such that for every $y, z \in B(0,q(x))$, it holds that

$$K^{-1}\rho(\Psi_x y, \Psi_x z) \le \|y - z\| \le A(x)\rho(\Psi_x y, \Psi_x z)$$

such that $e^{-\varepsilon} < \frac{A(f(x))}{A(x)} < e^{\varepsilon}$, where ρ is the distance on manifolds.

Proof. We prove in four steps. We will find $q(x)$ in three steps and prove (4) in the fourth step.

Step 1. We need a $q(x)$ so that (2) is satisfied. This candidate will be r_0.

Apply Theorem 2, for each $x \in \mathcal{R}$ consider the Lyapunov change of coordinates $C_\varepsilon(x)$ from T_xM to $\mathbb{R}^n$, and we get

$$A_\varepsilon(x) = C_\varepsilon^{-1}(f(x))\mathrm{d}_x f C_\varepsilon(x) \tag{6}$$

that has the Lyapunov block form after the Lyapunov change of coordinates.

For each $x \in M$ and $r > 0$, denote the disc of radius r centered at $x \in T_xM$ by $D_x^r M = \{w \in T_xM \mid \|w\| < r\}$.

Choose $r_0 > 0$ smaller than the injective radius so that $\|\mathrm{d}_w \exp_x\|$, $\|\mathrm{d}_{\exp_x w} \exp_x^{-1}\| \le 2$ for every $w \in D_x^{r_0} M$ and that $\exp_{f(x)}$ is injective on

$$\exp_{f(x)}^{-1} \circ f \circ \exp_x(D_x^{r_0} M).$$

The map the $f_x : C_\varepsilon(x)(D_x^{r_0} M) \to \mathbb{R}^n$ can be written as

$$f_x = \Psi_{f(x)}^{-1} \circ f \circ \Psi(x) = C_\varepsilon^{-1}(f(x)) \circ \exp_{f(x)}^{-1} \circ f \circ \exp_x \circ C_\varepsilon(x). \tag{7}$$

From Equation (6) and (7), we immediately see that $A_\varepsilon(x) = \mathrm{d}_0 f_x$. It is therefore of the Lyapunov block form, so (2) is true if $q(x) \le r_0$.

Note that the Euclidean ball $\{w \in T_xM \mid \|w\| < r_0 \|C_\varepsilon^{-1}(x)\|^{-1}\}$ is contained in $C_\varepsilon(x) D_x^{r_0} M$.

Step 2. Estimate the C^1 distance between f_x and $\mathrm{d}_0 f_x$ to find $q(x)$ so that it satisfies (3). The candidate will be $\delta(x)$.

We have two remainders on $\mathbb{R}^n$ and T_pM:

$$r_x(u) = f_x(u) - \mathrm{d}_0 f_x u,$$
$$g_x(w) = \exp_{f(x)}^{-1} \circ f \circ \exp_x - \mathrm{d}_x f w,$$

with $r_x(0) = 0$, $\mathrm{d}_0 r_x = 0$ and $\mathrm{d}_0 g_x = 0$. Since $f \in C^{1+\alpha}$, there is a constant $L > 0$ such that $\|\mathrm{d}_w g_x\| \le L\|w\|^\alpha$, so

$$\begin{aligned}\|\mathrm{d}_u r_x\| &= \|\mathrm{d}_u(C_\varepsilon(f(x)) \circ \exp_{f(x)}^{-1} \circ f \circ \exp_x \circ C_\varepsilon^{-1}(x))\| = \|\mathrm{d}_u(C_\varepsilon(f(x)) \circ g_x \circ C_\varepsilon^{-1}(x))\| \\ &\le L\|C_\varepsilon(f(x))\|\|C_\varepsilon^{-1}(x)\|\|C_\varepsilon^{-1}(x)u\|^\alpha \le L\|C_\varepsilon(f(x))\|\|C_\varepsilon^{-1}(x)\|^{1+\alpha}\|u\|^\alpha.\end{aligned}$$

In particular, for $\varepsilon > 0$, $\|u\| < \delta(x) = \varepsilon^{1/\alpha} L^{-1/\alpha} \|C_\varepsilon(f(x))\|^{-1/\alpha} \|C_\varepsilon^{-1}(x)\|^{-1-1/\alpha}$ implies $\|\mathrm{d}_u r_x\| < \varepsilon$.

(This is why we need $C^{1+\alpha}$.)

Step 3. We find $q(x)$ to satisfy (1).

We show $\delta(x)$ is tempered, so we can use Lemma 3.1 to get $q(x)$. This is true, since $\|C_\varepsilon(x)\|$ is tempered, we have

$$\lim_{m\to\pm\infty} \frac{1}{m} \log \delta^{\pm 1}(f^m(x)) = \pm \lim_{m\to+\infty} \frac{1}{m} \log \left(\varepsilon^{1/\alpha} L^{-1/\alpha} \|C_\varepsilon(f^{m+1}(x))\|^{-1/\alpha} \|C_\varepsilon^{-1}(f^m(x))\|^{-1-1/\alpha}\right) = 0.$$

Thus, we get $q^{-1}(x) \ge \delta^{-1}(x)$ that satisfies the tempering kernel condition. Clearly, $q(x) \le \delta(x)$ satisfies (1), (2), (3).

Step 4. Proof of Statement (4).

It follows from (2) that

$$\|y - z\| = \|\Psi_x^{-1}(\Psi_x(y)) - \Psi_x^{-1}(\Psi_x(z))\| \le 2\|C_\varepsilon^{-1}(x)\|\rho(\Psi_x(y), \Psi_x(z)).$$

We also have

$$\rho(\Psi_x(y), \Psi_x(z)) \le 2\|C_\varepsilon(x)\|\|y - z\| \le 2\|y - z\|.$$

By Lemma 3.1, we choose $K_\varepsilon(x) \ge \|C_\varepsilon^{-1}(x)\|$ such that $K_\varepsilon(x)e^{-\varepsilon} \le K_\varepsilon(f(x)) \le K_\varepsilon(x)e^{\varepsilon}$. We take $A(x) = 2K_\varepsilon(f(x))$ and $K = 2$, and the result follows. □

Remarks. For $x \in \mathcal{R}$, the set $R(x) = \Psi_x(B(0, q(x)))$ is called a **regular neighborhood** of x, or a **strong Lyapunov chart** at x.

We see from the statement of Theorem 3 that the way we construct the chart relies on the reduction theorem.

In statement (1), we want to find a tempered function $q(x)$. This is saying that the size of the regular neighborhood, whose size may not be invariant and may decrease along the trajectory, but the rate can be arbitrarily low.

Reference

[1]. Luís Barreira, Yakov Pesin, Introduction to Smooth Ergodic Theory: Second Edition, American Mathematical Society, 2023